

WASP-52b

R-0478

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_250810 · created 2026-07-20T05:45:17+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460897.7802 +/- 0.0091 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.202 +/- 0.062
Transit depth (Rp/Rs)^2	4.06 +/- 2.49 [%]
Orbital Inclination (inc)	83.68 +/- 2.79
Airmass coefficient 1 (a1)	4.64 +/- 3.24
Airmass coefficient 2 (a2)	-1.04 +/- 0.53
Scatter in the residuals of the lightcurve fit is	76.62 %
Best Comparison Star	#1 - [255, 198]
Optimal Aperture	0.75
Optimal Annulus	3.25
Transit Duration (day)	0.058 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.202 $\pm$ 0.062 vs 0.1646 (deviation 0.49 σ)
Duration fit vs published	0.058 d vs 0.0758 d (deviation 0.5 σ)
Mid-time O-C	-18.6 min
Depth SNR	2.7
Residual scatter	76.62 %

Light curve

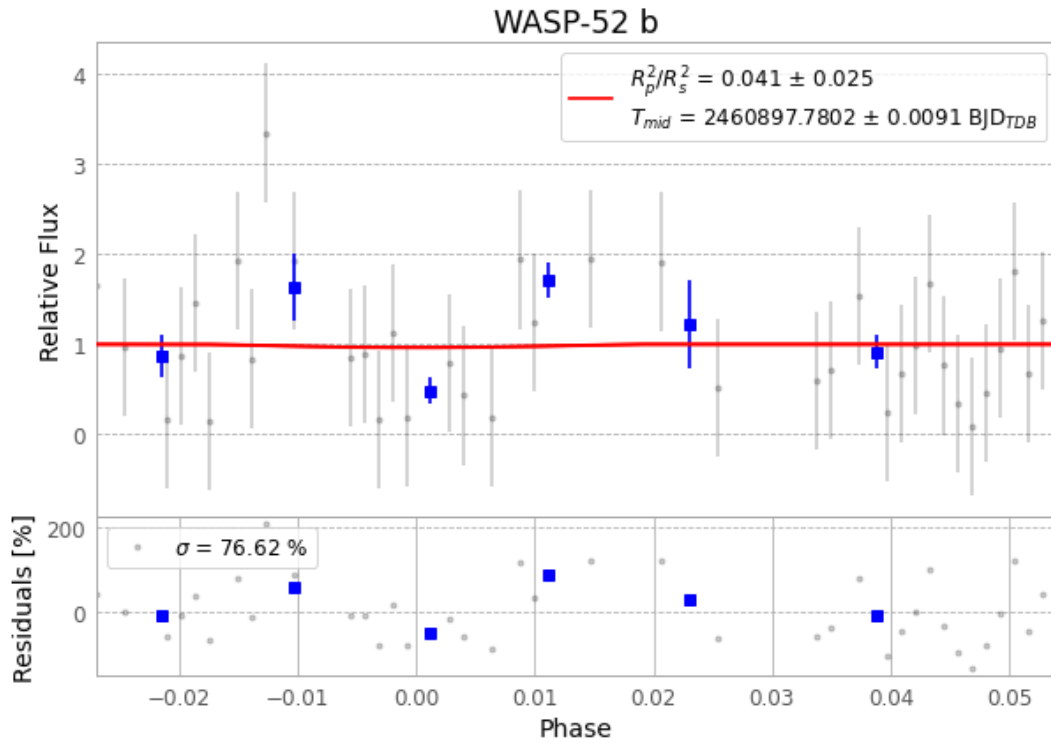


Figure 1 — FinalLightCurve_WASP-52 b_2025-08-09.png (SHA-256 d4bbdfe9a7bd5089...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [567, 415], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 73667c8758ba9194...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

69 FITS frames (45 MB), WASP-52250810052715.FITS → WASP-52250810085115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-250810.log (58a48c0fe0e0...), FinalLightCurve_WASP-52 b_2025-08-09.png (d4bbdfe9a7bd...), FinalLightCurve_WASP-52 b_2025-08-09.csv (bb854e885cbf...)

Compute resources

Wall time 165.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 05:45Z.